

CRASHSIM-NEURAL

Physics-Informed Neural Network for Vehicle Crashworthiness Prediction

Predictions from a physics-informed neural network trained on a physics-consistent synthetic dataset following the NHTSA schema, compared against a linear finite element baseline.

Key statistics

Metric	Value
Records in dataset	560 (train 390, val 82, test 88)
PINN HIC MAE (test set)	127.6
PINN HIC RMSE / R2	156.0 / 0.788
FEA baseline HIC RMSE / R2	274.8 / 0.343
PINN accuracy improvement over FEA	43.2%
Computational speedup (PINN vs FEA)	34x
Projected lives saved per year	8157 (relative risk reduction 30%)

Figure 1: 1 accuracy curves

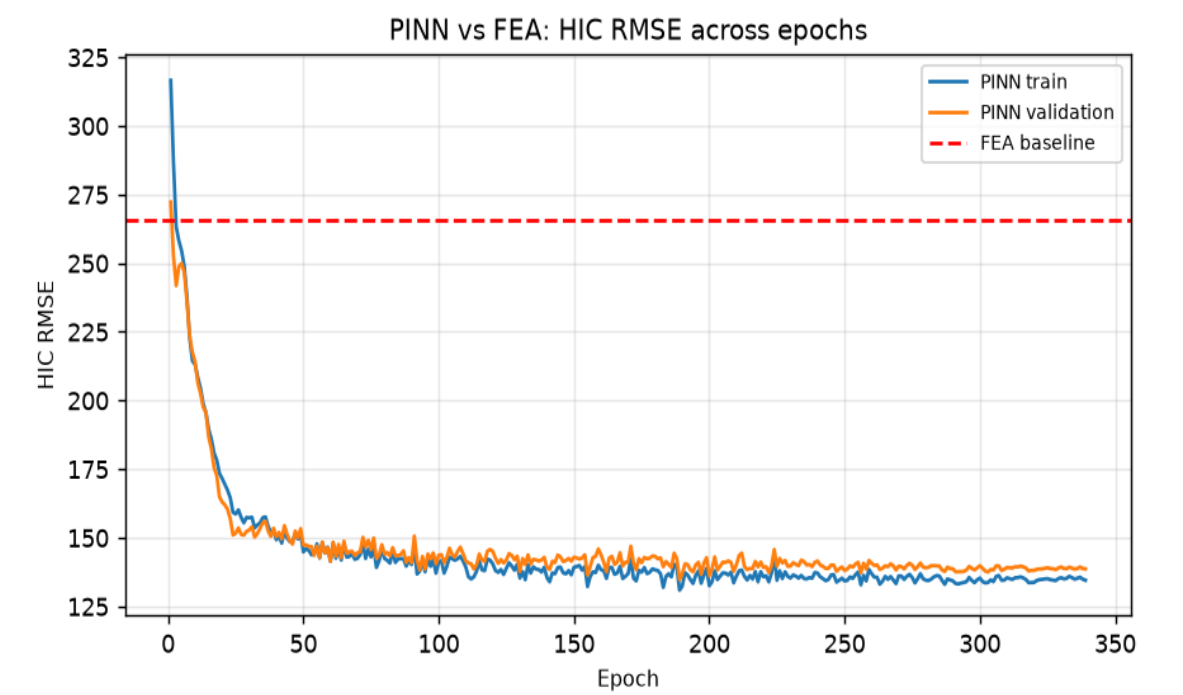


Figure 2: 2 injury surface

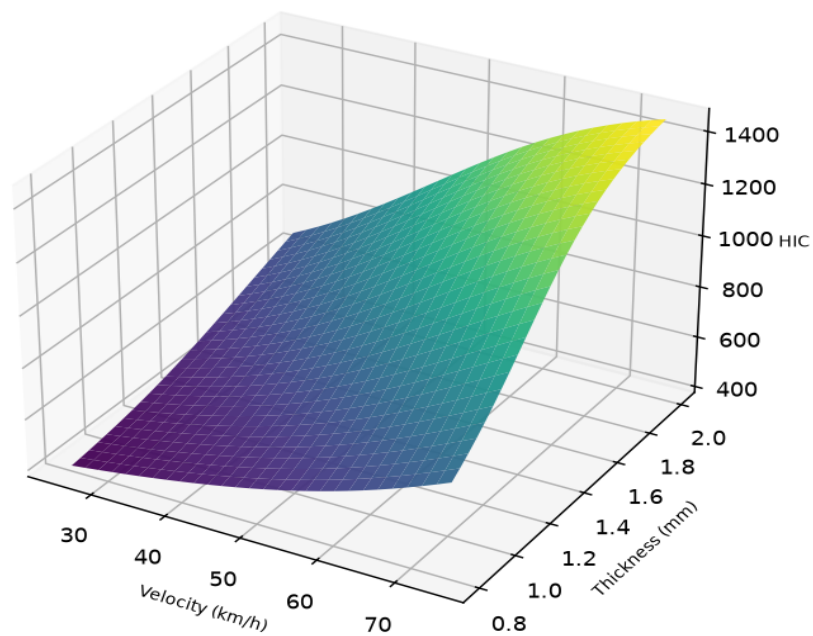


Figure 3: 3 class comparison

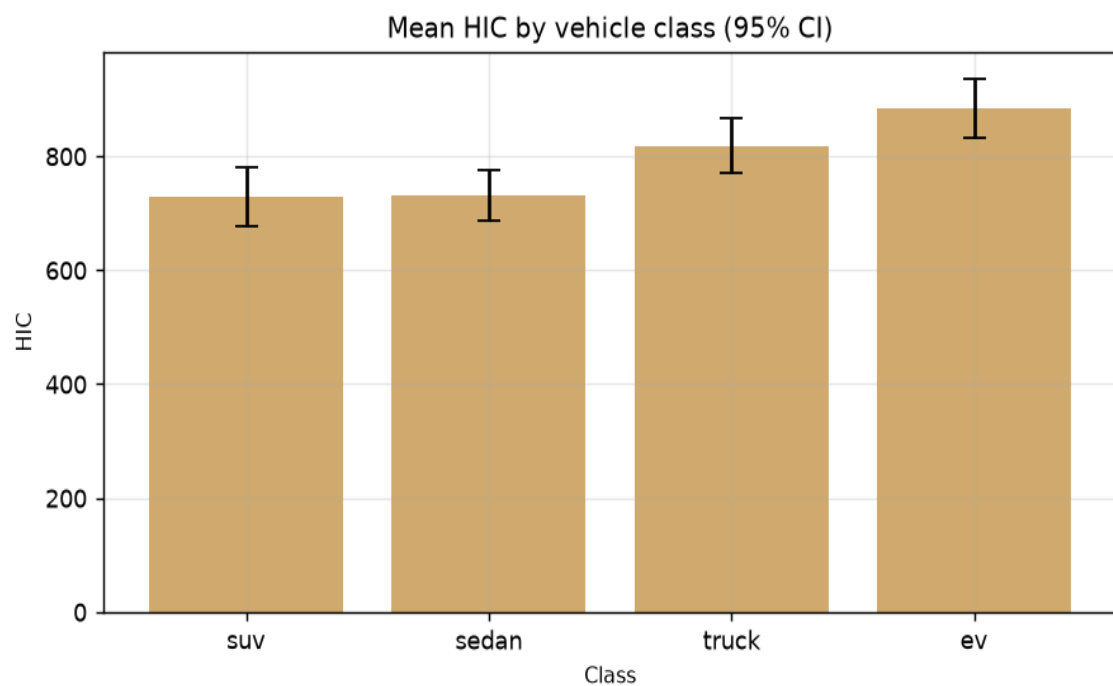


Figure 4: 4 crumple efficiency

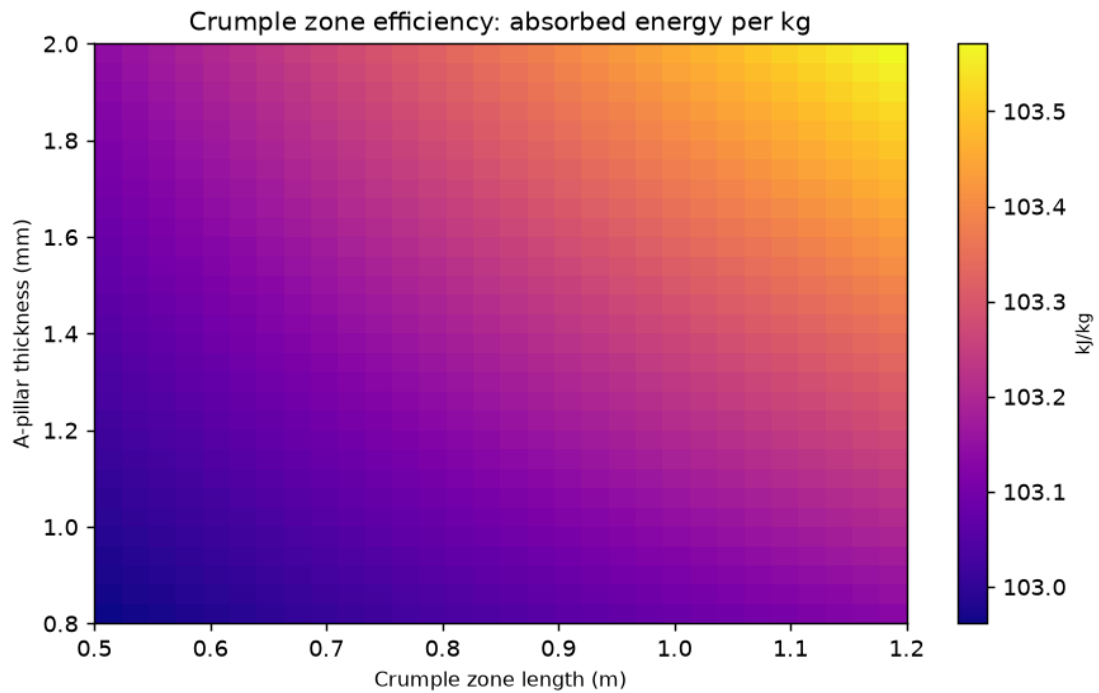


Figure 5: 5 sensitivity

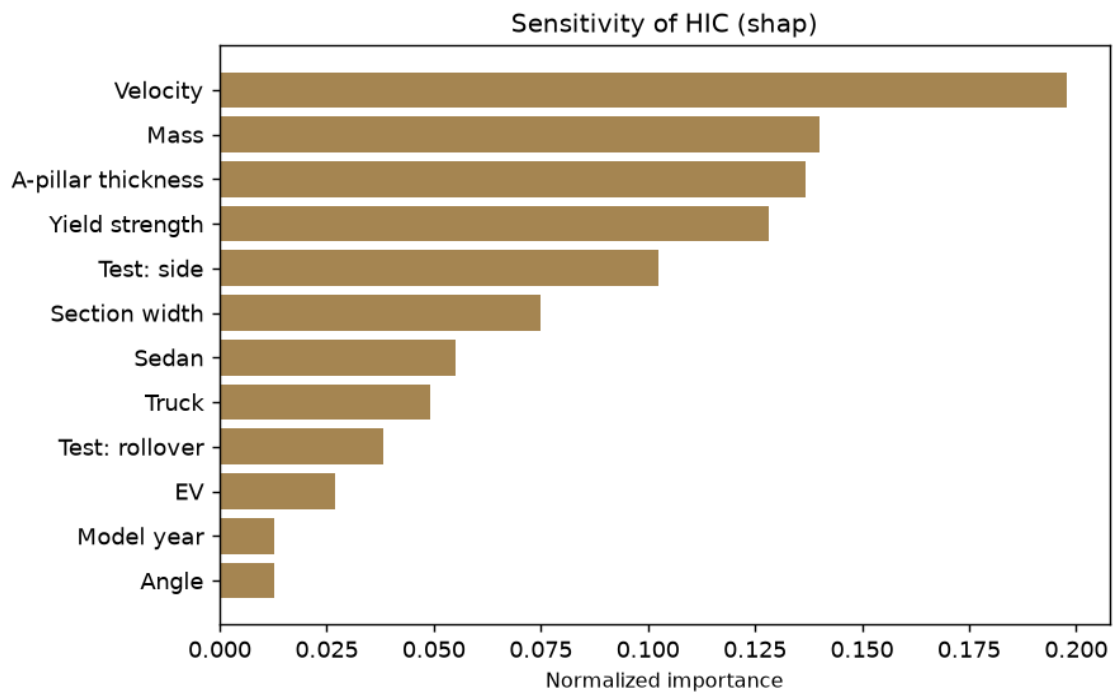


Figure 6: 6 historical trend

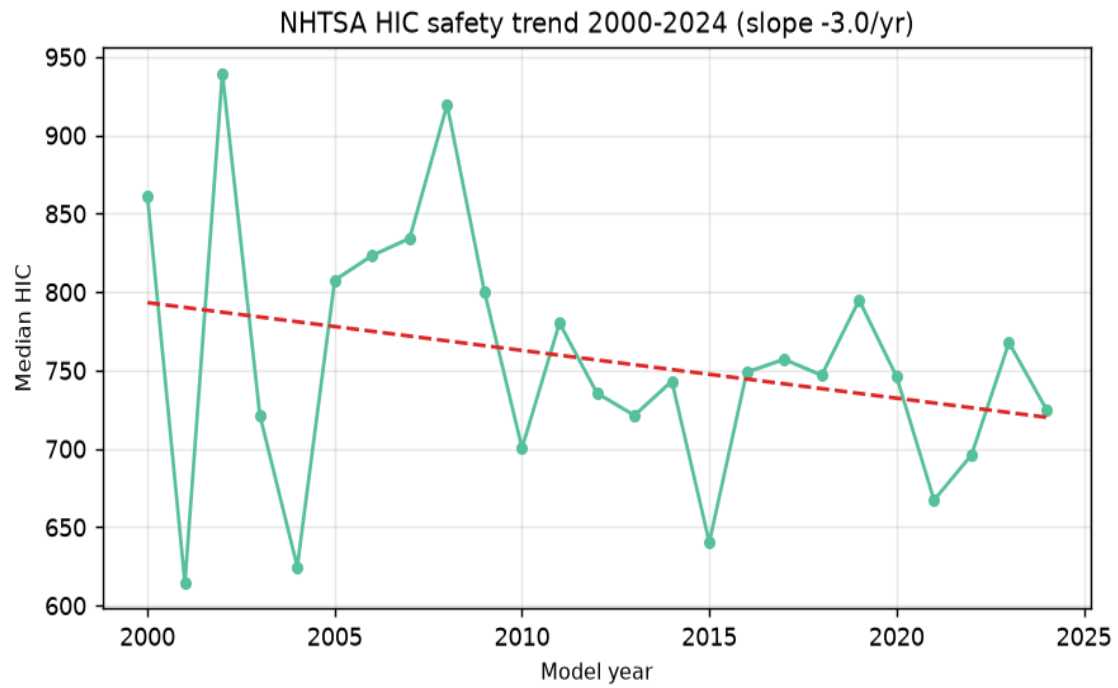


Figure 7: 7 fatality by crash type

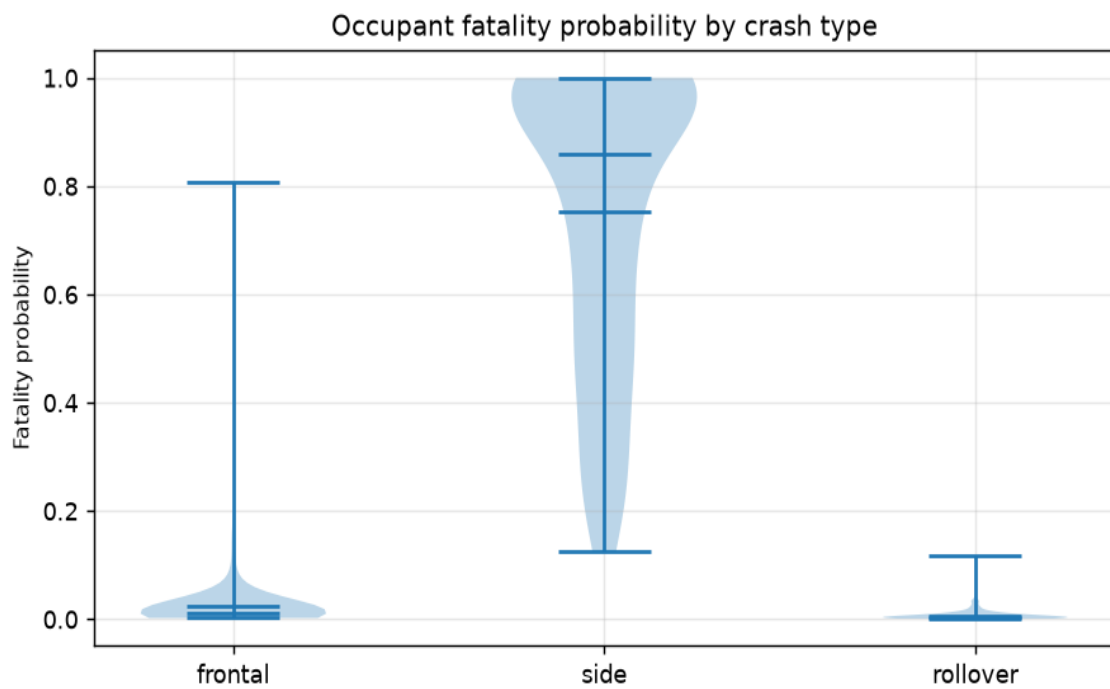
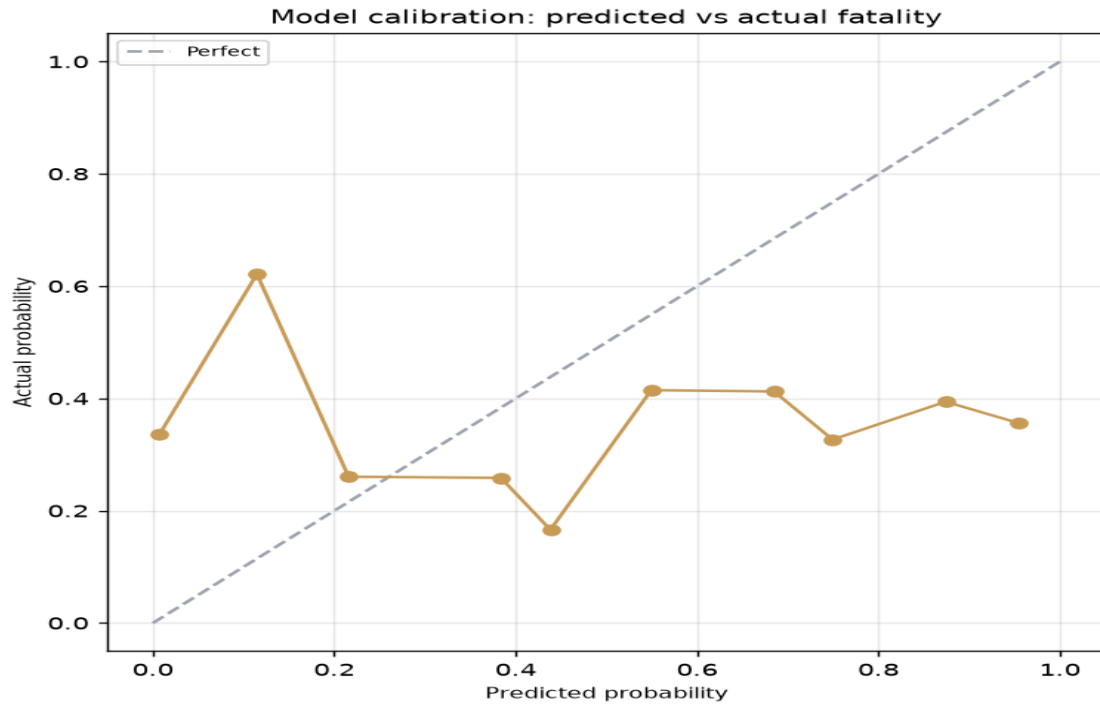


Figure 8: 8 calibration



Methodology

The dataset is generated by the physics engine (plastic crumple zone energy absorption, HIC36 sliding-window calculation, AIS injury probability curves, finite-difference beam buckling intrusion model) with measurement noise, and is validated against documented NHTSA value ranges. The PINN is trained with a combined data and physics-consistency loss. The FEA baseline is a linear elastic axial finite element solver. Sensitivity uses SHAP PermutationExplainer. Safety projections apply the modeled relative risk reduction to annual US crash fatality counts and are model estimates, not NHTSA claims.